Customer Shopping Behavior Analysis Report

Project Overview

This project analyzes customer shopping behavior using transactional data stored in a CSV file. The workflow includes data preprocessing in Python, storing the cleaned data in MySQL, executing SQL-based business analysis, and visualizing insights in Power BI dashboards.

Dataset Summary

- Source: CSV Dataset
- Rows : 3,900
- Columns : 18
- Main attributes :
    - customer demographics (Age, Gender, Location)
    - product details (Item Purchased, Category, Purchased Amount, Size, Color)
    - purchase behavior (Previous Purchased, Frequency of Purchases, Review Rating, Shipping Type)
    - Discounts (Discount Applied, Promo Code Used)
- Missing Data : 37 Values in Review Rating column

Data Preprocessing using Python

- **Data Loading :** Load the CSV dataset using pandas.
- **Dataset Structure and Data Types :** Used *df.info()* to check structure of data and *df.describe()* for summary statistics of numeric column.

```
[62]: # describe the statistics of numeric column
df.describe()
```

[62]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Handled missing Values :** Checked for missing values using *df.isnull().sum()* and filled missing entries in the **Review Rating** column with the median review rating of their respective **Category** using *groupby()* and *fillna().*

- **Column Standardization** Converted column names into a standardized lowercase underscore format for easier analysis and documentation.

- **Create Additional Derived Columns:**
    - Created an **age_group** derived column by segmenting customers into quartile-based age categories (Young Adult, Adult, Middle Aged, and Senior) using *pd.qcut()*.
    - Created a **purchase_frequency_days** derived column by mapping purchase frequency categories to their corresponding number of days using a dictionary-based transformation.

- **Database Integration :** Connected Python script to MYSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis using MySQL

Performed analytical SQL queries in MYSQL to solve major business problems :

1. **Revenue analysis by gender -** Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. **High-spending customers using discounts -** Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount	avg_purchase_amount
2	64	59.7644
3	73	59.7644
4	90	59.7644
7	85	59.7644
9	97	59.7644
12	68	59.7644
13	72	59.7644
16	81	59.7644
20	90	59.7644
22	62	59.7644
24	88	59.7644
29	94	59.7644
32	79	59.7644
33	67	59.7644
35	91	59.7644
37	69	59.7644
40	60	59.7644
41	76	59.7644
43	100	59.7644
44	69	59.7644
55	94	59.7644
57	73	59.7644
58	64	59.7644
60	79	59.7644
62	68	59.7644
64	79	59.7644
65	83	59.7644

3. **Top-rated products -** Found products with the highest average review ratings.

Result Grid | Filter Rows:

item_purchased	avg_rating
Handbag	3.8
Gloves	3.8
Shoes	3.8
Sandals	3.8
Sneakers	3.8

4. **Shipping Type Comparison –** Compared average purchase amounts between Standard and Express shipping.

Result Grid | Filter Rows: | Exp

shipping_type	ROUND(AVG(purchase_amount), 2)
Express	60.48
Standard	58.46

5. **Subscribers vs. Non-Subscribers comparison –** Compared average spend and total revenue across subscription status.

subscription_status	total_customer	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Top discounted products –** Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation –** Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	customer_count
Loyal	3116
Returning	701
New	83

8. **Top products by category -** Listed the most purchased products within each category.

rank_num	category	item_purchased	total_purchases
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions –** Checked whether customers with >5 purchases are more likely to subscribe.

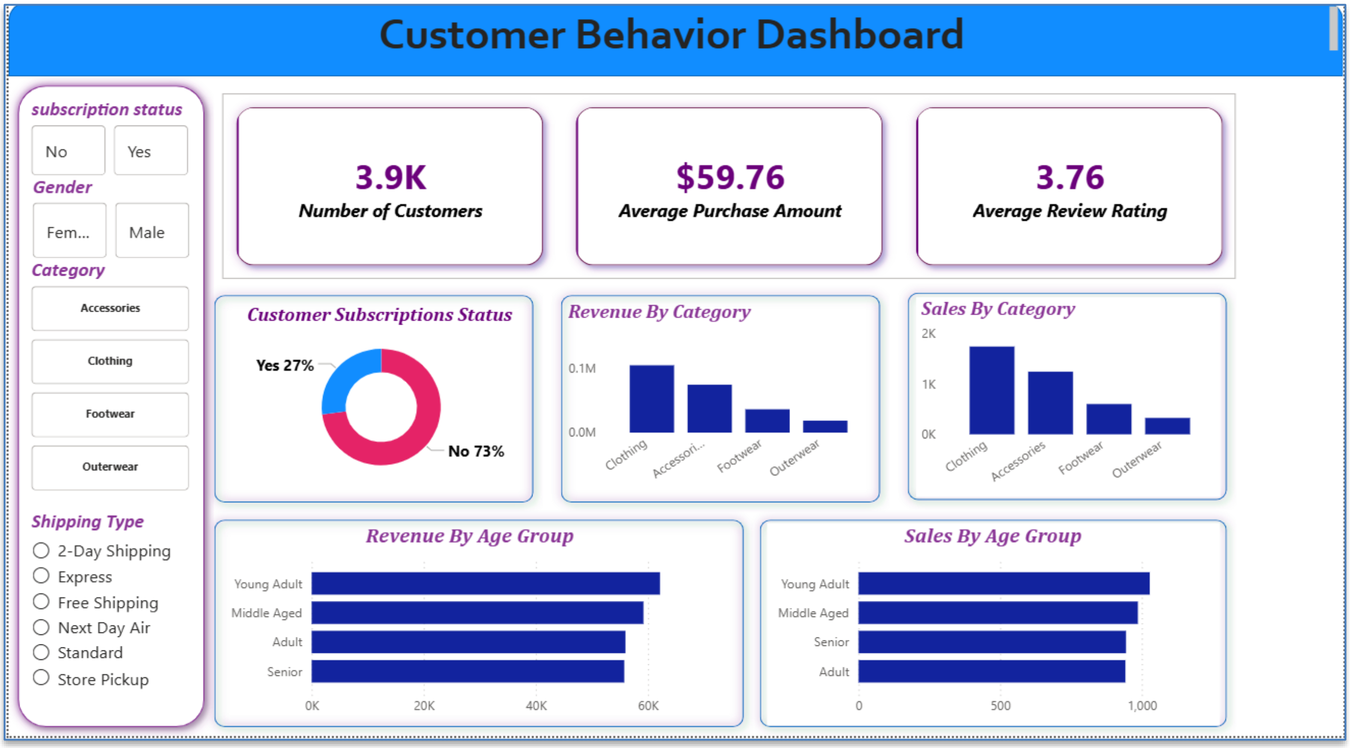
subscription_status	repeat_buyers
Yes	958
No	2518

10. **Revenue contribution by age group -** Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle Aged	59197
Adult	55978
Senior	55763

Dashboard Creation in Power BI

Connected MySQL database with Power BI and created interactive dashboards to visualize customer purchase behavior, product trends, and revenue insights.



Key Insights

- **Certain product categories generated higher revenue** – Some products were purchased more often and contributed more to total sales.
- **Discounts influenced purchase frequency** – Customers bought products more frequently when discounts were available.
- **Subscribers showed stronger purchase consistency** – Subscribers made purchases more regularly compared to non-subscribers.
- **Loyal customers contributed significantly to revenue** – Repeat customers generated a large portion of overall business revenue..

Business Recommendations

- **Improve customer loyalty programs** – Reward customers to encourage repeat purchases and long-term engagement.
- **Promote subscription benefits** – Offer special benefits to attract more subscribers and increase retention.
- **Optimize discount strategies** – Use discounts wisely to increase sales while maintaining profit.

- **Focus marketing on high-value customer segments –** Target loyal and frequent buyers with personalized marketing campaigns.

Conclusion

This project demonstrates how Python, MySQL, and Power BI can be integrated to transform raw customer shopping data into meaningful business insights.